Legal Issues of Generative AI Training: Copyright and Data Protection Under EU Law

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Abstract— **Generative AI systems train on vast collections of copyrighted works and personal data scraped from the internet, creating serious tensions with both EU copyright and data protection law. This paper examines whether the existing legal frameworks can adequately handle these challenges. On the copyright side, we argue that the text and data mining exceptions introduced by the CDSM Directive were designed for analytical research and do not comfortably extend to commercial systems that generate competing creative content. The opt-out mechanism offers no real solution, shifting the entire burden onto rights holders while leaving creators uncompensated. On the data protection side, the GDPR's requirements for lawful processing, transparency and cross-border transfers collapse when applied to the mass, indiscriminate scraping that AI training requires. Data subject rights exist on paper but become practically unenforceable at this scale. What emerges is a clear pattern: Europe is attempting to apply legal frameworks built for different technological realities to a problem they were never designed to solve. Purpose-built regulation for AI training is not optional but highly necessary.**

I. Introduction

With the arrival of AI, the world has encountered a new legal crisis. Systems like ChatGPT, Midjourney and many others are now able to produce text, images and creative content that truly rivals human work. There is a catch though: to create such models, companies have used vast collections of existing works, such as books, articles, photographs and art, most of which were extracted from the internet without permission.

Multiple legal problems arise from this. Two of the most important ones are the following: firstly, copying millions of copyrighted works in order to train AI models almost certainly violates copyright law, unless an exception applies. The second problem is that these training datasets usually contain personal information about real people, which raises serious questions under European data protection law. Both issues remain largely unresolved to this day, leaving AI developers, creators and regulators in a state of considerable uncertainty.

The matter is, European law was not built for something like this. The EU's 2019 copyright reforms introduced narrow exceptions for "text and data mining", which were designed for academic researchers analyzing data and not for commercial companies that use them to train systems generating competing content. Meanwhile, the GDPR imposes strict rules on transferring personal data internationally, rules that assume orderly business relationships and contractual safeguards. Web scraping at the scale required for AI training does not fit comfortably within either framework.

This paper tackles both problems head-on. We start by explaining how generative AI works and why it is not just "fancy data analysis". We then examine whether EU copyright and data protection laws can adequately handle AI training — and spoiler alert, they struggle. What becomes clear is that we are trying to apply old legal tools to new technological problems, and the fit so far is quite poor. Europe needs purpose-built regulation for AI, not improvisation with laws designed for different challenges.

II. Technical Background

II-A How Generative AI Works

Generative AI systems are different in that they work through machine learning. More specifically, rather than programming every instruction, we teach computers to recognize patterns by showing them many examples to "learn" from. These systems use artificial neural networks, loosely inspired by the way our brains wire neurons together.

The way these systems actually learn is by learning from their mistakes. The system makes an attempt, observes whether it was successful or not, calculates how wrong it was and adjusts its memory accordingly. After repeating this millions of times with enough data — with enough data meaning billions of examples — the system becomes capable of creating new content that resembles what it studied. Following this process, text models learn natural word sequences, image models learn how visual elements work together, and even music models absorb patterns in melody and rhythm.

The important part of this process though is the following: it requires making copies of everything used for training. All those books, images and songs get downloaded, converted into formats the computer can read, stored on the company's servers, cleaned up and accessed repeatedly during the learning process. Even after the training phase has finished, some pieces of that original content can get baked into the model itself. Researchers call this process "memorization". Sometimes it is even possible to prompt these systems just right and they will extract nearly exact copies of things they saw during training.

II-B The Critical Distinction: TDM vs. Generative AI

And this is precisely where the legal problems arise. Traditional text and data mining is all about analysis. Researchers use it to spot trends, count how often words appear, find correlations and even generate statistics. The result is always analytical knowledge. Whether it is a paper explaining which topics tend to cluster together or a chart mapping sentiment patterns, text and data mining (TDM) extracts facts about the data without mimicking or copying its creative expression.

Systems that leverage Generative AI on the other hand don't analyze content to produce insights. They produce new creative content by drawing on previously seen data. The output isn't a research finding — it is a new poem, a completely unknown illustration, a song never heard before. When a generative model writes a fantasy novel, it is not reporting on fantasy novels it studied. Instead, it produces a new fantasy novel that competes in the same market.

For copyright law, this distinction is critical. TDM exceptions exist because analytical research cannot replace or compete with the original work being studied. Generative AI is the complete opposite. A model trained on millions of random photographs can mass-produce new stock photos, potentially cutting into the market for real photographers. A model trained on journalism can produce news and articles that directly compete with actual reporters and their work.

This technical difference creates a legal puzzle: should exceptions written for analytical systems also cover creative systems?

III. EU Legal Framework

III-A Copyright Law

The foundation of European copyright law is the Berne Convention of 1886. That convention established something important: creative works deserve protection and authors should be the only ones deciding how their work gets used. The EU updated these principles in 2001 through the Information Society Directive [4], which hands copyright holders control over reproductions of their work. That protection creates a wide protective net. Making a digital copy? Protected. Downloading and converting to another format? Protected. Even storing a temporary version in computer memory? Still protected.

Copyright imposed by the law does not mean unlimited control though. Copying is not always wrong. Research, education and other uses sometimes need exceptions. These exceptions though come with tight limits. Article 5(5) sets up a three-part requirement: exceptions only work in narrow situations, cannot undermine how creators actually make money and cannot unreasonably damage their interests. Courts cannot simply expand these exceptions whenever new technology appears. The limits are real and they do matter.

III-B CDSM Directive TDM Exceptions

In 2019, the EU added two new exceptions specifically for TDM [3]. Article 2(2) defines what TDM actually means: "any automated analytical technique aimed at analyzing text and data in digital form in order to generate information which includes, but is not limited to, patterns, trends, and correlations." The key word there is "information". The law envisions extracting facts and insights, not creating new creative works.

Article 3 gives a narrow exception for researchers and cultural institutions doing scientific work. Article 4 becomes broader, allowing anyone to mine lawfully accessible content for any purpose. That's where things get complicated, though. Copyright owners have the option to prevent this by choosing to opt out. They just need to reserve their rights "in an appropriate manner", like using machine-readable signals for online content.

Why does this matter? Because when legislators wrote these rules in 2017-2019, they were thinking about academics studying disease patterns or analyzing historical texts. Generative AI as we know it today barely existed. The official impact assessments discuss research application like bioinformatics and rare disease studies. Nobody was contemplating systems that would generate competing novels or stock photos. The exceptions were written for one world, and we are now living in a different one.

III-C EU AI Act

The 2024 AI Act [2] finally tackles artificial intelligence head-on, though it does not solve the copyright mess. Article 53 tells companies making general-purpose AI models that they need copyright compliance policies and must respect opt-outs under the CDSM Directive's Article 4. Recital 105 acknowledges that these TDM techniques get used extensively when training AI on copyrighted material.

Here is the problem though. The AI Act demands transparency without answering the fundamental legal question. Companies must publish summaries of what they trained on, but these summaries explicitly cannot identify specific works. How is a photographer supposed to check if their images were used when summary just says "millions of web images"? Even more basically, the AI Act simply assumes Article 4 provides legal cover for training. It doesn't create new exceptions or clarify whether Generative AI actually counts as legitimate text and data mining.

III-D GDPR

The General Data Protection Regulation [1] kicks in whenever AI training touches personal data. And it almost always does. Scrape the internet for training data and you inevitably capture people's names, faces, email addresses, location information, and countless other personal details.

The GDPR sets out principles that AI training finds hard to meet. Article 5 demands that data processing be lawful, fair, transparent, limited to specific purposes, minimized to what is necessary, and accountable. Article 6 requires a legal basis before you can process personal data at all. Articles 13 and 14 say you must tell people when you are using their data. And Chapter V, perhaps most problematically, imposes strict rules on sending personal data outside the EU. You need either an official adequacy decision saying the destination country has equivalent protections, appropriate safeguards like Standard Contractual Clauses, or you must fit into narrow exceptional circumstances.

These rules were written for a world where businesses knowingly collect data from identifiable people and transfer it to specific recipients under clear contracts. AI training doesn't work that way. Companies scrape billions of web pages without knowing whose data they're grabbing, where those people are located, or where the data ends up during training that might span data centers across continents. The law expects order and transparency. AI training operates through mass collection and distributed processing.

IV. Copyright Challenges

IV-A Prima Facie Infringement in AI Training

Training AI means copying on an enormous scale. Companies scrape millions of works from the internet and store them on their servers. Converting these files into AI-readable formats creates yet another set of copies. Data gets stored redundantly. During training, systems access these copies repeatedly, loading portions into memory thousands of times. Every step reproduces copyrighted works. Under EU law, reproduction is the copyright holder's exclusive right. The Information Society Directive [4] is explicit: owners control reproduction "in any manner or form," whether permanent or temporary, complete or partial. Making copies without permission is prima facie infringement.

What happens after training makes things worse. Model weights may encode copyrighted expression through "memorization". Models sometimes reproduce near-exact copies of training examples. Large language models output entire paragraphs verbatim. Image models regenerate recognizable photographs. Researchers have developed techniques to extract training data from deployed models.

Copyrighted expression gets embedded in the model itself, distributed to millions of users, reproducible on demand. This resembles creating derivative works, making permanent copies, and enabling mass distribution-all exclusive rights belonging to copyright holders.

Scale matters. Traditional disputes involve one work or perhaps dozens. AI training involves billions simultaneously. A single model might train on essentially the entire accessible internet. The scope of potential infringement is unprecedented in copyright history.

Unless an exception applies, this violates copyright law. The question becomes: does Article 4's TDM exception cover generative AI training?

IV-B GenAI Training Qualify as TDM? The Core Debate

This gets us to the heart of the legal debate: can you call generative AI training "text and data mining" under Article 4 [3]?

AI companies believe you can. OpenAI, Stability AI, and others have built their entire approach around Article 4 providing legal permission. From their perspective, the reasoning is obvious: automated analysis of digital content is exactly what TDM means. The exception allows mining "for any purpose," so training generative models should count. And since lawmakers didn't explicitly exclude generative AI, why wouldn't it apply?

But look closely at the legal text and problems emerge. Article 2(2) defines TDM as techniques "aimed at analysing text and data in digital form in order to generate information." That word "information" matters. The definition continues: "information which includes but is not limited to patterns, trends and correlations." These are analytical outputs. Facts extracted from data. Not creative content that competes with the original works.

Generative AI does something fundamentally different. Sure, it analyzes patterns during training. But what comes out the other end isn't information about those patterns. It's new creative works built using those patterns. When Midjourney produces an image, it's not telling you facts about the photographs it studied. It's making a new photograph. That's synthesis, not analysis [7], [8].

The legislative history backs this up. Parliament wrote these provisions in 2017-2019 with a specific goal: letting researchers and businesses extract knowledge from data. The impact assessments talk about analyzing medical research, studying how language works, improving search engines [7]. All of these extract facts. None of them generate competing creative content. Generative AI as we know it today barely existed when lawmakers debated these rules.

Then there's the three-step test problem. Even if you somehow fit generative AI training into the TDM definition, you still have to satisfy Article 5(5) of the InfoSoc Directive [4]. Exceptions can't conflict with how works are normally exploited or unreasonably harm rightsholders. Training models that produce competing content arguably fails both requirements.Take a stock photography company that trains an AI on millions of stock photos, then starts selling AI-generated alternatives at lower prices. That's direct competition with the original market. Hard to claim that doesn't conflict with normal exploitation.Some legal scholars push for reading Article 4 more narrowly [7], [8]. They note that the CDSM Directive's preamble emphasizes innovation and research, not commercial content generation.The European Copyright Society suggests that Article 4 should only cover text and data mining (TDM) activities that are used for their intended research purpose, and not for activities that could directly compete in the market. Courts haven't resolved this definitively yet, though cases are moving through the system. But the textual evidence, the legislative history, and the purpose behind the law all point the same way: Article 4 was designed for analysis. Generative AI is about creation. The gap between them isn't just technical. It's fundamental.

IV-C The Opt-Out Mechanism's Structural Failures

Article 4's opt-out mechanism sounds reasonable in theory. Copyright holders can reserve their rights "in an appropriate manner, such as machine-readable means" for content made publicly available online. Problem solved, right? Creators who don't want their work used for AI training can just opt out.

Except it doesn't work. Not at the scale AI training operates. Start with the practical reality. AI companies have already trained their models on billions of works scraped from the internet over the years. Those works are already embedded in existing models. An opt-out placed today does nothing about yesterday's copying. The mechanism is purely prospective. It can't undo what's already been done.

Then there's the discovery problem. How is a photographer supposed to know their images were scraped from some website and used for training? Article 53 of the AI Act [2] requires companies to publish summaries of training data, but these summaries explicitly cannot identify individual works. A summary saying "millions of images from the internet" tells creators nothing useful. You can't opt out of something when you don't know it's happening.

The technical implementation is also a mess. There's no standardized protocol for machine-readable opt-outs. Some proposals suggest using metadata tags or robots.txt files, but these aren't universal and AI companies aren't necessarily respecting them. Different companies implement different systems, forcing creators to opt out separately for each AI developer. Good luck tracking them all down.

Scale makes everything worse. A professional photographer might have tens of thousands of images distributed across multiple platforms. A writer might have hundreds of articles on dozens of websites. Each one would need individual opt-out signals. For most creators, this is simply impossible to manage.

The opt-out mechanism shifts the entire burden onto creators. Instead of requiring AI companies to get permission before using works, the law requires creators to actively prevent use. For a technology that operates by scraping everything it can find, this gets the incentives exactly backward.

IV-D Comparative Perspective: US Fair Use

The US takes a completely different approach. American law doesn't have neat TDM exceptions. Instead, what the Supreme Court calls an "equitable rule of reason." You can use copyrighted works without permission for criticism, research, teaching and similar purposes. Courts will look at four things: your purpose, the type of work, how much you used and whether you're hurting the market.

The key concept is "transformative use." Does your new work add something original, or does it just replace what you copied? For years, courts said highly transformative uses get a pass, even commercial ones. The Supreme Court said in Campbell v. Acuff-Rose that transformative works further copyright's basic goal, which is to promote creativity.

But then came Warhol v. Goldsmith [6] in 2023. The Court ruled that adding artistic meaning isn't enough by itself. Justification is also necessary, especially when the new use serves the same commercial purpose as the original work. Warhol's Prince silkscreen failed as fair use when sold to illustrate a magazine, because that's exactly how the original photograph was used. Same artwork in a museum? Might be fine. Context matters.

So where does this leave AI training? Multiple billion-dollar lawsuits are testing the question right now. Authors sued over their books. American courts will decide whether training counts as transformative use or just mass copying.

The transatlantic split creates real problems. AI companies face completely different rules depending on which side of the ocean they're on.

IV-E The Remuneration Gap

Even if you accept that Article 4 covers AI training the fact that nobody pays for anything is not addressed. AI companies use millions of copyrighted works commercially, generating billions in revenue, while creators receive nothing.

That's not how copyright normally works. Radio stations pay royalties. Streaming services license content. But Article 4's opt-out framework creates what the EU describes as a "mandatory, fee-free exception." Unlike other copyright exceptions—such as private copying, which pairs exceptions with levies to ensure compensation—Article 4 imposes no duty to remunerate authors when their works get repurposed for training [7].

Creators aren't happy. In 2023, European authors and performers submitted an open letter to the Commission calling for "built-in safeguards to ensure authors are neither deprived of income nor stripped of control" when their works feed AI training [7]. Their proposals include collective licensing schemes, revenue-sharing agreements, and creating a dedicated remuneration right for AI training uses.

Several scholars have proposed solutions [7], [8]. These models draw on established EU practices like cable retransmission rights, public lending rights, and private copying levies.

News publishers make a pointed comparison: Article 15 of the CDSM Directive [3] gave them neighboring rights specifically because digital platforms were extracting value without paying. Why shouldn't the same logic apply to AI [7]?

The remuneration gap reveals a policy choice. Copyright has always worked on a simple principle: if you're making serious money from someone else's creative work, you pay them. Article 4 throws that out the window. It lets companies systematically exploit copyrighted material commercially without any payment obligation. Is this smart innovation policy or just a wealth transfer from artists to tech companies? That depends on who you ask.

V. Data Protection Challenges

V-A GDPR Applicability

The GDPR [1] applies whenever AI training involves personal data, which it almost always does. Training datasets scraped from the internet capture names, faces, emails and countless other personal details about real people.

AI companies need a legal basis under Article 6 to process this data. The most plausible options are consent (which they don't have), legitimate interests (debatable when processing billions of people's data), or public interest (hard to argue for commercial AI). Most companies seem to rely on legitimate interests, but that requires balancing their interests against individuals' rights. Given the scale and opacity of AI training, that balance is questionable [9].

Articles 13 and 14 require informing people when you process their data. How do you notify billions of internet users that you scraped their information? You don't. The GDPR assumes you know whose data you're processing. AI training breaks that assumption entirely [9].

V-B Cross-Border Data Transfers

Chapter V of the GDPR demands "an essentially equivalent level of protection" when personal data leaves the EU [1]. AI training fails at nearly every step of the required compliance process.

Without an adequacy decision, you need Article 46 safeguards like Standard Contractual Clauses. But SCCs require identifying both exporter and importer and having a contract between them. Web scraping billions of pages means no identifiable importer exists and no contract can be formed [11].

The Schrems II judgment [5] makes things worse. The CJEU ruled that transfers must be "suspended or stopped" where equivalent protection cannot be guaranteed. Following Schrems II, the EDPB requires Transfer Impact Assessments examining "the law and practices in force in the third country" to ensure they don't impinge on safeguards [11]. How do you conduct Transfer Impact Assessments for indiscriminate web scraping? You don't really know whose data you're collecting, which EU member states they're in or where training will occur until it starts across multiple data centers. The EDPB's six-step roadmap assumes orderly, documented flows between identified parties [11]. AI training operates through mass scraping without documentation, contracts or identified recipients. Chapter V's entire framework collapses.

V-C Data Subject Rights

The GDPR gives individuals the right to access, correct and delete their personal data in accordance with Articles 15, 16, and 17 [1]. Training artificial intelligence makes it nearly impossible to exercise these rights. How can individuals request the erasure of their data when they do not know that it has been collected? How can controllers identify which training examples relate to specific individuals across billions of web pages that have been collected [9]?

Even if individuals could identify their data, technical challenges remain. Removing individual data from huge training datasets creates significant computational difficulties [12]. Furthermore, personal data may remain in the model weights through memorization, meaning that deletion from the training datasets provides only a superficial solution [9]. The GDPR warns that supervisory authorities may need to order the deletion of entire datasets or models when personal data has been unlawfully processed [11].

The fundamental mismatch is clear: data protection legislation presupposes the enforcement of individual rights, but artificial intelligence training operates on a global scale. Rights exist on paper, but in practice they are inapplicable.

V-D Regulatory Guidance

Although fragmentation remains, regulators have started to provide guidance. For example, the Swedish Data Protection Authority's guidelines of February 2025 stress that among others, legitimate interests as a legal basis require careful weighing, adequate documentation and compliance with the data minimisation principles [10].

Quite similarly, the EDPB's Opinion 28/2024 of December 2024 addresses data protection aspects when training AI models [11]. That opinion clarifies that legitimate interests must be assessed on a case-by-case basis, substantiated from the outset and not contrary to law. The revised EDPB Guidelines on Generative Artificial Intelligence of January 2025 further provide practical guidance on how to exercise data subjects' rights and implement appropriate measures [12].

However, guidelines alone cannot resolve the fundamental tensions. Regulators recognise that the scale and opacity of AI training create challenges that their frameworks struggle to address.

VI. Conclusions

Generative AI has created a legal crisis that Europe's existing frameworks cannot adequately resolve. The technology operates by training on vast collections of copyrighted works and personal data scraped from the internet, creating tensions with both copyright and data protection law that lawmakers did not anticipate.

On copyright, the core problem is Article 4's TDM exception [3]. The text, legislative history and purpose all suggest it was designed for analytical research extracting facts and patterns, not commercial systems generating competing creative content [7], [8]. AI companies have proceeded as though Article 4 provides blanket authorization, but this interpretation stretches the exception far beyond its intended scope. The opt-out mechanism offers no real solution. It operates purely prospectively, requires creators to discover uses they cannot detect and shifts the entire burden onto rights holders. Meanwhile, creators receive no compensation despite AI companies generating billions in revenue from their works.

Data protection law faces similar difficulties. The GDPR [1] requires controllers to know whose data they process and can inform individuals accordingly, while AI training breaks these assumptions entirely [9]. Companies scrape billions of web pages without even knowing whose personal data they capture or where those people are located. Legal bases prove elusive, transparency obligations become impossible, and cross-border transfer rules collapse when applied to indiscriminate web scraping [5], [11]. Data subject rights exist on paper but become unenforceable at the scale AI operates.

Both frameworks share the same fundamental problem: they were built for different technological realities. Copyright law envisions specific works being copied for identified purposes. Data protection law envisions orderly processing where controllers maintain relationships with data subjects. Generative AI training operates through mass appropriation and distributed processing that does not quite fit either model. Europe needs purpose-built regulation for AI training, not improvisation with laws designed for different challenges. The current approach leaves everyone in regulatory limbo. Until lawmakers address these issues directly, the legal uncertainty will persist.

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